

Laravel + React + Inertia Core Concept

Local Development Flow

Browser Request

Browser ↓ Nginx ↓ Laravel Route ↓ Inertia::render('Home') ↓ app.jsx loads Home.jsx ↓ React renders page

```
---  
  
# Main Files Responsibility  
  
## app.blade.php  
  
Purpose:  
  
Starts React application
```

NOT frontend UI.

Equivalent to:

React index.html

```
Contains:  
  
- Vite loader  
- Inertia root  
- React mounting point  
  
---  
  
## app.jsx  
  
Purpose:  
  
Loads React pages dynamically
```

Main React/Inertia bootstrap file.

Pages/

Purpose:

```text id="gqk5wx" Actual screen pages

Examples:

```
```text id="j4f8yf"
Login.jsx
Dashboard.jsx
Home.jsx
```

Layouts/

Purpose:

```text id="6w1qux" Common page structure

Examples:

- sidebar
- navbar
- footer
- wrappers

---

## Components/

Purpose:

```
```text id="j9n6uv"
Reusable UI pieces
```

Examples:

- Button
- Input
- Modal
- Card

Local Development

Uses:

```
```bash id="yhn8gh" npm run dev
```

Vite dev server runs continuously.

Purpose:

- hot reload
- fast refresh
- instant frontend updates

Ports:

Service	Port
nginx	8000
vite	5173
mysql	3306

---

# Production Flow

Production does NOT use:

```
```text id="axvk4u"
npm run dev
```

Instead:

```
```bash id="vh1eq8" npm run build
```

creates compiled frontend files inside:

```
```text id="s1ij44"
public/build
```

Now Laravel directly serves compiled frontend assets.

No running Vite server needed.

Production Request Flow

```text id="p0zq3n" Internet ↓ Nginx ↓ Laravel ↓ Compiled React frontend

```

What Goes Live On Server

Main live files:

```text id="1fj1wm"
app.blade.php
app.jsx
Pages/
Layouts/
Components/
routes/
controllers/
models/
```

Compiled frontend generated into:

```text id="7jlwm4" public/build

```
during:

```bash id="iv3j1wm"
npm run build
```

Simple Interview/Explanation Version

Laravel

Handles:

- backend
- authentication

- routing
 - sessions
 - database
-

React

Handles:

- frontend UI
 - dynamic pages
 - user interaction
-

Inertia

Acts as bridge between:

```
text id="psjlm"
```

```
Laravel ↔ React
```

without creating separate API architecture.

Core Concept

app.blade.php

Starts React app.

app.jsx

Loads requested page.

Page

Controls screen content.

Layout

Controls page structure.

Component

Reusable small UI block.